

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 27

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines (1) • Also possible on the flood when the sea is higher than the basin; no generation when levels are equal (slack water) (1)	2
01.2	<i>1 mark for each point.</i> • Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow; the timing does not match demand (1) • Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned (1)	2
01.3	<i>1 mark for each point.</i> • Spring tides: Sun and Moon aligned so their gravitational effects combine giving a large tidal range; neap: effects oppose giving a small range (1) • Energy depends on the mass of water and height difference, so output is much greater at springs (1)	2
01.4	<i>1 mark for each point.</i> • When demand is low, surplus electricity pumps water into the lagoon above sea level (1) • The stored water is released through the turbines later when demand is high (pumped storage) (1)	2
01.5	<i>1 mark for each point.</i> • Large tidal range (eg Severn Estuary) (1) • Estuaries, bays or headlands / channels that focus the flow and increase velocity (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
02.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Stress-related illness: ulcers, high blood pressure, heart disease (1) • Behavioural changes: irritability, aggression, lack of concentration leading to accidents (1) • Communication problems / educational underperformance (eg schools under flight paths) (1) • Sleep disturbance (1)	3
02.2	<i>1 mark for each point.</i> • Acoustic fatigue: objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) (1) • Vibration damage: repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes (1) • Shock impacts: the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry (1)	3
02.3	<i>1 mark for each point.</i> • Birds that sing to establish territories / attract mates cannot be heard against background noise (1) • Birds scared by sudden noise leave nests so predators take eggs / chicks; cliff-nesting seabirds knock eggs or chicks into the sea (1)	2
02.4	<i>1 mark for each point.</i> • Whales / dolphins use hearing / echolocation to find food and communicate; loud sonar causes hearing damage if close to the source (1) • Behavioural changes / disorientation possibly linked to mass strandings; reduced feeding success (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Deforestation: less interception so less evaporation from leaves; less transpiration so less water returned to the atmosphere and less precipitation downwind; more infiltration and surface runoff, higher river flows / flooding (2) • Urban development: impermeable concrete / tarmac reduce infiltration and aquifer recharge; runoff is faster and greater, increasing river flooding downstream; drains carry water quickly to rivers (2) • Agriculture: soil compaction by machinery / trampling reduces infiltration and soil moisture, increases runoff; irrigation increases evaporation; loss of soil biota (worms) reduces infiltration and water retention; drainage lowers water tables (2)	6
03.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Higher temperatures increase evaporation and change the amount, timing, type (rain vs snow) and location of precipitation (1) • More rapid melting of snow and ice changes river regimes: more flooding after rain, lower flows in dry periods (1) • Changed wind patterns move humid air to different areas (1) • Sea level rise / salt incursion into coastal aquifers (1)	2
03.3	<i>1 mark for each point.</i> • Most water abstracted is cleaned and returned to the river / sea so the main impact is on quality (pollution) not quantity (1) • Effects are noticeable only where large urban centres draw on restricted local supplies (aquifers, small rivers) (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 116-117; AQA spec 3.2.2.1.	
04.1	<i>1 mark for each point.</i> - Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well (1) - Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well (1) - Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate (1)	3
04.2	<i>1 mark for each point.</i> - Natural gas may enter aquifer water (1) - Injected chemicals may reach aquifers or the surface; toxic metals in the rock may be mobilised (1) - Large volumes of water needed (1) - Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally) (1) - Reduced by: collection, treatment and re-use of waste water; well casing standards; restricting sites away from aquifers / sensitive areas; monitoring (1)	5
04.3	<i>1 mark for each point.</i> - Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices (1) - Reserves only include what is economically exploitable now; high environmental impacts (land take, water, CO₂) also limit exploitation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1.	
05.1	<i>1 mark for each point.</i> - Inorganic mercury sank into the sediments / deep water where conditions are anaerobic / low oxygen (1) - It was chemically reduced / converted by (anaerobic) bacteria to methyl mercury (1)	2
05.2	<i>1 mark for each point.</i> - Methyl mercury is liposoluble so it passes through cell membranes and is stored in fat / not excreted (1) - So it bioaccumulates in organisms and biomagnifies along the food chain (eg in fish, then humans eating fish) (1) - It is a neurotoxin / teratogen causing brain damage and birth abnormalities (eg Minamata) (1)	3
05.3	<i>1 mark for each point.</i> - Mercury is persistent / not degraded; it is an element so cannot be broken down (1) - It remains in the sediments and continues to be released / converted and taken up by the food chain; long-lived fish continue to accumulate it (1)	2

Question	Marking guidance	Marks
05.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Water current velocity and direction (dispersal) (1) • Properties of the pollutant: solubility, persistence, liposolubility / potential to bioaccumulate (1) • Food chain pathways to humans (fish, shellfish) / the critical group (1) • Sediment type / adsorption onto clay (1) • Volume of the water body / residence time / dilution (1) • Sensitive locations downstream (fisheries, drinking water abstraction) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 238-242; AQA spec 3.4.1, 3.4.2.	
06.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Discharge of tank-washing water from tankers (1) • Oil rig / drilling accidents / leakage during drilling (1) • Pipeline leaks (1) • Leakage from storage tanks / refinery spills (1) • Waste lubricating oil from vehicles / industry washed into drains (1) • Other ship accidents (fuel oil) (1)	3
06.2	<i>1 mark for each point.</i> • Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia (1) • Feathers lose buoyancy / waterproofing so heavily oiled birds drown (1) • Birds swallow toxic oil when preening; time spent cleaning feathers reduces time for feeding themselves / their young (1)	3
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen (1) • Toxic components (benzene, xylene) dissolve and poison plankton near the surface (1) • Oil blocks light so algae / phytoplankton cannot photosynthesise (1) • Oil masks odours / narcotises so animals cannot find mates or food by smell (1) • Shore organisms (molluscs, crustaceans) are covered and asphyxiated (1)	2
06.4	<i>1 mark for each point.</i> • Dispersal / dilution by currents and waves is prevented so oil stays concentrated; in open water most spills disperse in days or weeks (1) • Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
07.1	<i>Correct answer only.</i> • B - Low liposolubility and low persistence (1)	1
07.2	<i>1 mark for each point.</i> • Bioaccumulation: build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown (1) • Biomagnification: increase in concentration of the pollutant at successive trophic levels of a food chain (1)	2

Question	Marking guidance	Marks
07.3	1 mark for each point. • They are intended to be toxic (to the target species) (1) • They are deliberately released / spread into the environment (rather than being an accidental or waste release) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
08.1	Correct answer only. • B - rivers are accessible over a large area with few natural contaminants (1)	1
08.2	1 mark for each point. • Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) (1) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals (1)	2
08.3	1 mark for each point. • Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) (1) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
09.1	1 mark for the correct answer. • $(2.9 - 0.7) / 2.9 \times 100 = 75.9\%$ (accept 76%) (1)	1
09.2	1 mark for each point. • Although the larger vehicle uses more energy in total, it carries many more people / a larger load (1) • So the energy per passenger (or per tonne of goods) is lower; steel wheels on rails also have low rolling resistance (1)	2
09.3	Any 3 from the points below - 1 mark each, maximum 3. • Flexibility of routes / destinations and door-to-door convenience (1) • Time / speed and reliability where public transport is infrequent (1) • Low fuel tax / cheap fuel (1) • Carrying goods; comfort; status (1)	3
09.4	1 mark for each point. • Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey (1) • So more journeys shift from cars to bulk transport / cycling (1) • ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely (1) • Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	
10.1	1 mark for each point. • t-test (or Mann-Whitney U if not normally distributed) (1) • Comparing two means of counted / measured data from two independent samples (1)	2

Question	Marking guidance	Marks
10.2	<i>1 mark for each point.</i> - The difference is statistically significant (less than 1% probability that it arose by chance) (1) - The null hypothesis (no difference between fields) is rejected; the organic field has significantly more earthworms (1)	2
10.3	<i>1 mark for each point.</i> - A significant difference does not show a causal relationship (1) - Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference (1)	2
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Scientific methodology: identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research - Sample location: random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias; interval chosen so variation is detected without excess data - Sample timing: diurnal, seasonal, weather-related and long-term changes; sampling on different occasions; interval matched to rate of change - Sample size and number: larger and more samples for non-homogeneous variables; running mean and standard deviation show when the mean is reliable; preliminary study to decide - Standardisation and control of variables; accuracy, precision, reliability, representativeness; anomalies and repeats; calibration of equipment; appropriate techniques and risk management - Statistical analysis: choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation - Evaluation: trade-offs between cost, time, disturbance and reliability; limitations of sub-sampling; justified conclusion	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	A valid investigation begins with a clear, testable hypothesis and a null hypothesis, informed by existing knowledge, so that the data collected can actually accept or reject it. The first design decision is sample location: random sampling (grid coordinates or numbered sites) or systematic sampling along transects avoids the bias that would come from choosing convenient sites, and a preliminary study shows the interval needed to detect variation without collecting unnecessary data.	
	Timing matters because many variables change diurnally, seasonally or with weather, so samples on different occasions are needed and the interval must match the rate of change. Sample size and number depend on how variable the population is: a running mean shows when enough samples have been taken and the standard deviation shows the scatter around the mean. Techniques must be standardised so results from different sites, times or researchers are comparable, other variables controlled or monitored, and equipment calibrated so that precise readings are also accurate; anomalies should be repeated rather than ignored.	
	Finally the analysis must be chosen to suit the data - Spearman's rank for correlation, chi-squared for frequencies, a t-test for two means or Mann-Whitney U for medians - so that significance ($p < 0.05$) can be assessed, remembering that a significant correlation does not prove causation. Every decision trades reliability against cost, time and disturbance; sub-samples can only estimate the whole, so the most valid study is the one whose sampling design makes its limitations explicit.	
	Weaker answer - Level 2 (5 marks)	
	When planning fieldwork you need a hypothesis. You should choose sites randomly so it is not biased, for example using random numbers. You need to take enough samples and repeat them so the results are reliable. The method should be the same each time. You should measure accurately with calibrated equipment. Then you do a statistical test such as a t-test to see if the results are significant. If p is less than 0.05 the result is significant. This makes the conclusions valid.	
	What makes the difference	
	• Level 2 is sound but generic: it says 'enough samples' where Level 3 explains how a running mean and standard deviation tell you how many.	
	• Level 3 covers timing and the choice between statistical tests; Level 2 mentions one test only.	
	• The evaluation is missing from Level 2 - there is no discussion of trade-offs or of correlation vs causation, so it cannot reach the top level however accurate it is.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25